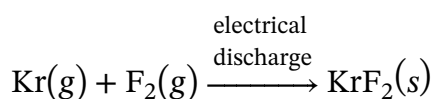
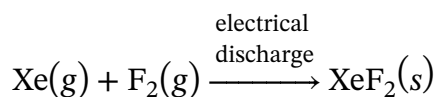


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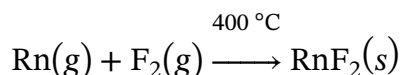


Reaction 1



Reaction 2

A similar reaction can be achieved in a mixture of $F_2(g)$ and $Rn(g)$ at an elevated temperature.



Reaction 3

The compounds formed from Reactions 1 and 2 are relatively stable and can be stored if shielded from water, heat, and light. Although the $RnF_2(s)$ from Reaction 3 is chemically stable, it still degrades over time because radon is radioactive and undergoes nuclear decay to become isotopes of Po, Bi, and Pb. In contrast, no stable fluorine compounds form when mixing $F_2(g)$ with $He(g)$, $Ne(g)$, or $Ar(g)$.

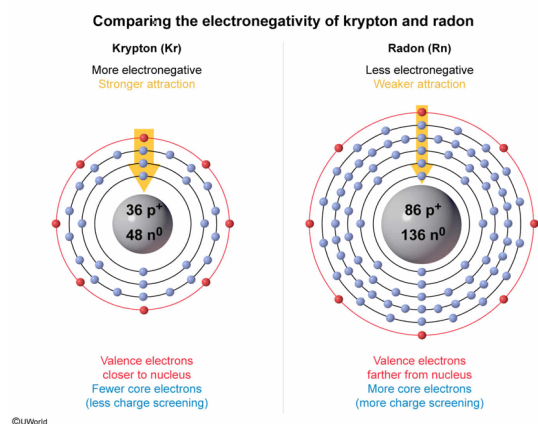
In the compounds formed in Reactions 1 and 3, the noble gas with the greatest electronegativity is:

- ☐ A. Kr, because it has a greater ratio of protons to neutrons in its nucleus. [8%]
- ✓ ☒ B. Kr, because it has fewer protons but its valence electrons are closer to its nucleus. [83%]
- ☐ C. Rn, because it has a greater total number of electrons. [4%]
- ☐ D. Rn, because it has a greater number of protons in its nucleus. [4%]

Correct

83% answered correctly

Explanation:



Electronegativity is the tendency of an atom to **attract electrons within a bond**. The electronegativity of an atom depends on the distance of the **valence electrons** from the nucleus (ie, **Coulomb's law**) and on how well the **core electrons** shield the valence electrons from the nucleus (ie, the **effective nuclear charge**). On the **periodic table**, **electronegativity** tends to increase from left to right across a period (row) and from bottom to top in a group (column).

Applying the trends in electronegativity to compare the Kr and Rn atoms, Kr is higher in the Group 18 column and has a greater electronegativity than Rn. Although Kr has fewer protons (ie, less charge) in its nucleus, the valence electrons of Kr are much closer to the nucleus (ie, the valence electrons experience a stronger attraction).

(Choice A) Neutrons have no charge, and the ratio of protons (p^+) to neutrons (n^0) does not impact the effective nuclear charge of the nucleus. As such, this ratio also has no effect on the electronegativity of an atom.

(Choices C and D) Although Rn atoms do have more protons in the nucleus (ie, greater charge), the valence electrons are in a higher level farther from the nucleus. The greater number of total electrons in Rn atoms also means that more core electrons are positioned between the nucleus and the valence electrons (ie, more screening of the nuclear charge). These factors *decrease* the attraction of the valence electrons.

Educational objective:

Electronegativity is the tendency of an atom to attract electrons within a bond. The electronegativity of an atom depends on its effective nuclear charge and the distance of its valence electrons from that charge. On the periodic table, electronegativity tends to increase from left to right across a period (row) and from bottom to top in a group (column).

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Subject:

General Chemistry

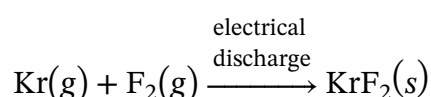
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System:

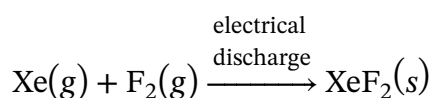
Atomic Theory and Chemical Composition

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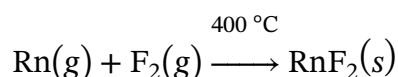


Reaction 1



Reaction 2

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Reaction 3

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If the chemical bond length is estimated using the sum of the atomic radii of the bonded atoms, which of the following compounds has the shortest bond with fluorine?

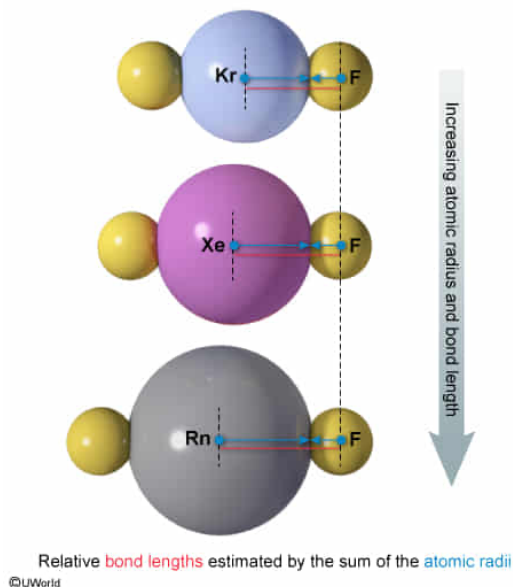
- ✔ ☒ A. $KrF_2(s)$ [84%]
- ☐ B. $XeF_2(s)$ [1%]
- ☐ C. $RnF_2(s)$ [12%]
- ☐ D. The relative bond lengths of the compounds cannot be determined without more information. [1%]

Correct

83% answered correctly

Explanation:

Comparing bond lengths of noble gas compounds



The **atomic radius** of an atom is the distance from the center of its nucleus to the outermost shell containing the valence electrons. By considering **atomic radius trends** on the periodic table, the relative sizes of elemental atoms can be compared based on their positions on the table.

Within each group (column), the atomic radius generally **increases from top to bottom**. Each square down the column **adds another shell** of electrons farther from the nucleus. The increased number of **core electrons** also shield the outer **valence electrons** from the attraction of the protons in the nucleus, which causes the outer shell of electrons to remain positioned farther from the nucleus.

Within each period (row), the atomic radius tends to **decrease from left to right** (with some exceptions). Each square to the right adds one proton to the nucleus and one electron to the outer shell, but the total number of shells remains constant across the row. As a result, the added proton increases the Z_{eff} , which causes the outer electrons to be attracted more strongly and pulls the electron shells closer to the nucleus.

The chemical bond length can be estimated as the sum of the atomic radii of the bonded atoms. In KrF_2 , XeF_2 , and RnF_2 , the contribution from the F atom is the same in each. As such, the lengths of the Kr–F, Xe–F, and Rn–F bonds can be compared by considering the relative sizes of the noble gas atoms. Because Kr is the farthest up the Group 18 column, it has the smallest atomic radius and the Kr–F bond is the shortest of the given bonds.

(Choices B, C, and D) The atomic radii of the noble gases increase down the column. As such, bonds between these atoms and fluorine increase in length as the atomic radius of the noble gas atom increases.

Educational objective:

On the periodic table, the atomic radius of atoms tends to decrease from left to right across a

row and increase from the top to the bottom of a column. The length of a chemical bond can be estimated as the sum of the atomic radii of the bonded atoms.

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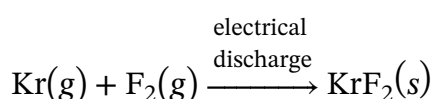
Subject:
General Chemistry

QID:403326

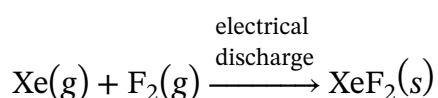
System:
Atomic Theory and Chemical Composition

Noble gases refer to the elements of Group 18 on the periodic table. These gases are commonly used in "neon lighting" because electric current passed through a sealed glass tube containing a noble gas causes the emission of photons of visible light at specific wavelengths (ie, colors) unique to the gas that is used.

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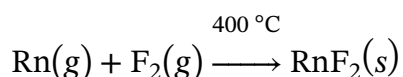


Reaction 1



Reaction 2

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Reaction 3

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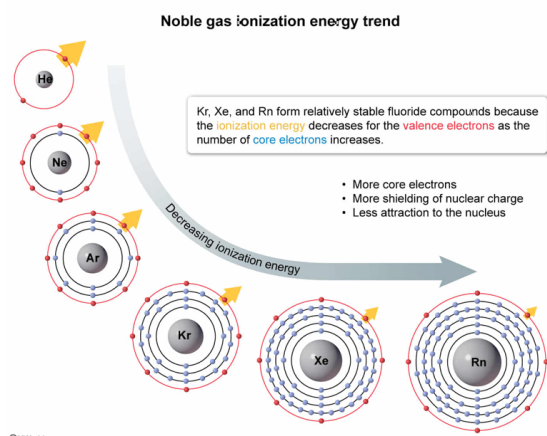
Which of the following best explains why Kr, Xe, and Rn form relatively stable fluoride compounds but He, Ne, and Ar do not?

- ☐ A. Electronegativity increases as the total number of electrons increases. [5%]
- ☐ B. Polarizability decreases as the size of an atom increases. [15%]
- ☐ C. Electron affinity increases as the number of electron energy levels increases. [17%]
- ✓ ☒ D. Ionization energy decreases for the valence electrons as the number of core electrons increases. [61%]

Correct

60% answered correctly

Explanation:



Each proton in the nucleus of an atom contributes one unit of positive charge that exerts an **electrostatic attraction** on the negatively charged electrons around the atom. However, in atoms with several electrons, the **core electrons** positioned between the **valence electrons** and the nucleus provide a **shielding constant S** that counteracts part of the full **nuclear charge Z** attracting the valence electrons. As a result, the valence electrons experience an **effective nuclear charge Z_{eff}** that is less than the full nuclear charge, such that:

$$Z_{\text{eff}} = Z - S$$

The precise value of S depends on the electron density of a given electron orbital configuration.

As stated in the passage, the noble gases form bonds by losing electrons (ie, ionization). The **ionization energy** of a valence electron in an atom is influenced by the effective nuclear charge of the atom. Larger values of Z_{eff} create a stronger attraction that takes more energy to overcome. Consequently, atoms with more core electrons (ie, more charge screening) have a lower Z_{eff} , making it easier to remove a valence electron.

Therefore, Kr, Xe, and Rn can form relatively stable fluoride compounds whereas He, Ne, and Ar do not, because the **ionization energy decreases** for the valence electrons as the number of core electrons increases down the Group 18 column.

(Choice A) **Electronegativity** is the tendency of an atom to attract electrons within a bond. This attraction does not depend solely on the total number of electrons, and electronegativity *decreases* down a column on the periodic table.

(Choice B) **Polarizability** *increases* as the **size of an atom** increases.

(Choice C) **Electron affinity** assesses the tendency of an atom to accept an additional electron, but the noble gases have nearly zero electron affinity and form the compounds described in the passage by *losing* electrons. In addition, electron affinity tends to *decrease* down a column of the periodic table

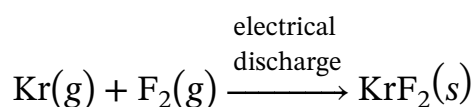
Educational objective:

The effective nuclear charge (Z_{eff}) of an atom depends on the number of protons in the nucleus and the number of core electrons providing charge screening around the nucleus. As the number of core electrons increases, Z_{eff} decreases and the ionization energy of the valence electrons decreases.

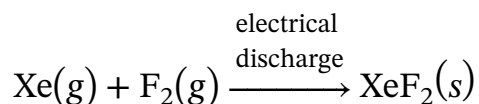
Time spent:0 sec**Subject:**
General Chemistry**QID:**403327**System:**
Atomic Theory and Chemical Composition

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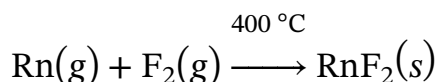


Reaction 1



Reaction 2

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Reaction 3

The compounds formed from Reactions 1 and 2 are relatively stable and can be stored if shielded from water, heat, and light. Although the $RnF_2(s)$ from Reaction 3 is chemically stable, it still degrades over time because radon is radioactive and undergoes nuclear decay to become isotopes of Po, Bi, and Pb. In contrast, no stable fluorine compounds form when mixing $F_2(g)$ with $He(g)$, $Ne(g)$, or $Ar(g)$.

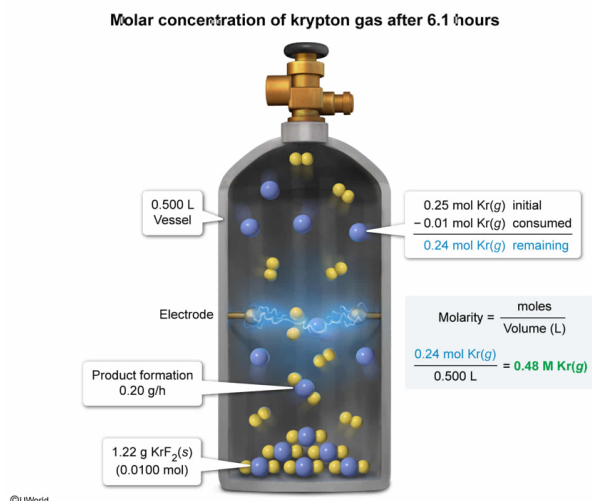
Suppose that Reaction 1 is performed in a sealed 500 mL vessel containing 0.25 mol $Kr(g)$ and 0.25 mol $F_2(g)$, resulting in the formation of $KrF_2(s)$ at an average rate of 0.20 grams per hour. What is the approximate molar concentration of $Kr(g)$ remaining in the vessel after 6.1 hours? (Note: The molar mass of KrF_2 is 121.80 g/mol.)

- ☐ A. 0.010 M [15%]
☐ B. 0.24 M [34%]
☒ C. 0.48 M [41%]
☐ D. 0.50 M [8%]

Correct

41% answered correctly

Explanation:



If $\text{KrF}_2(\text{s})$ was formed at an average rate of 0.20 grams per hour during the reaction, multiplying this rate by the reaction time gives the approximate mass of $\text{KrF}_2(\text{s})$ produced:

$$0.20 \frac{\text{g KrF}_2(\text{s})}{\text{h}} \times 6.1 \text{ h} = 1.22 \text{ g KrF}_2(\text{s})$$

Dividing the mass of $\text{KrF}_2(\text{s})$ by its **molar mass** gives the approximate number of moles of KrF_2 :

$$1.22 \text{ g KrF}_2(\text{s}) \times \frac{1 \text{ mol KrF}_2(\text{s})}{121.80 \text{ g KrF}_2(\text{s})} = 0.0100 \text{ mol KrF}_2(\text{s}) \text{ formed}$$

Applying the **mole ratio** from Reaction 1 shows that an equal number of moles of $\text{Kr}(\text{g})$ was consumed during the reaction:

$$0.0100 \text{ mol KrF}_2(\text{s}) \times \frac{1 \text{ mol Kr}(\text{g})}{1 \text{ mol KrF}_2(\text{s})} = 0.0100 \text{ mol Kr}(\text{g}) \text{ consumed}$$

Subtracting the moles of $\text{Kr}(\text{g})$ used in the reaction from the number of moles of $\text{Kr}(\text{g})$ initially present in the vessel gives the number of unreacted moles of $\text{Kr}(\text{g})$ remaining

after the reaction:

$$0.25 \text{ mol Kr(g) initial} - 0.0100 \text{ mol Kr(g) consumed} = 0.24 \text{ mol Kr(g) remaining}$$

Dividing the unreacted moles of Kr(g) by the volume of the vessel (converted to liters) gives the **molar concentration (molarity)** of the unreacted Kr(g) remaining in the vessel after the reaction:

$$\text{Molarity} = \frac{\text{moles}}{\text{volume(L)}} = \frac{0.24 \text{ mol Kr(g)}}{0.500 \text{ L}} = 0.48 \text{ M Kr(g)}$$

(Choice A) 0.010 *moles* of Kr(g) reacted, but this is not the *molarity* (ie, moles per liter) of the *unreacted* Kr(g) remaining in the vessel.

(Choice B) 0.24 *moles* of unreacted Kr(g) remain after the reaction, but this is not the *molar concentration* (ie, moles *per liter*) in the vessel.

(Choice D) 0.50 M is the *initial* molar concentration of Kr(g) before the reaction begins. As such, the molar concentration remaining after 6.1 hours of the reaction must be less than the initial concentration.

Educational objective:

The molar concentration (ie, molarity) of a chemical species is defined as the moles of species per liter of mixture. The moles of a species can be found by dividing its mass by its molar mass. Mole ratios from a balanced chemical reaction can be used to relate the moles of one species to the moles of another species.

Time spent:0 sec

Subject:

General Chemistry

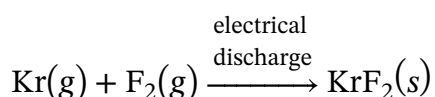
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System:

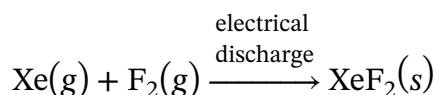
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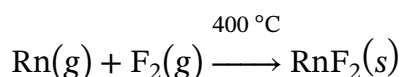


Reaction 1



Reaction 2

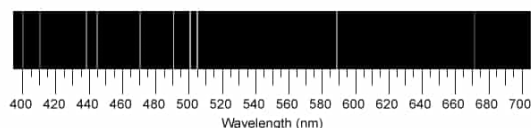
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Electric current passed through a sealed glass tube containing helium produces the emission line spectrum shown below.



Which of the following statements is true about the emission lines of this spectrum?

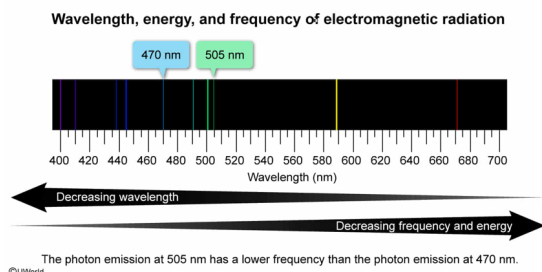
- ☐ A. The photon emission at 672 nm has higher energy than the photon emission at 589 nm. [7%]

- ✓ ☒ B. The photon emission at 505 nm has a lower frequency than the photon emission at 470 nm. [78%]
- ☐ C. The energy of the photon emission at 410 nm is inversely proportional to the photon frequency. [8%]
- ☐ D. The energy of the photon emission at 445 nm is directly proportional to the wavelength. [5%]

Correct

78% answered correctly

Explanation:



Electrons within an atom can exist only in particular, discrete energy levels identified by **quantum numbers** and **orbital** designations. An **electron excited** from a lower to a higher energy level must absorb an amount of energy equal to the energy difference between the two levels. Conversely, an electron moving from a higher to a lower energy level releases energy in the form of a photon (ie, a small packet of **electromagnetic radiation**) that has an energy equal to the energy difference between the levels.

Because each element has a unique **electron configuration**, excited electrons in a gas-phase sample of a pure element emit photons only at specific wavelengths rather than as a continuous spectrum. This produces a **line spectrum** with **emission lines** at **particular wavelengths** that are characteristic of the given element and its electron configuration. As a result, the line spectrum of each element is unique.

On the **electromagnetic spectrum**, the **energy of a photon E** is **directly proportional** to its **frequency ν** and **inversely proportional** to its wavelength λ , as given by the equation:

$$E = h\nu = \frac{hc}{\lambda}$$

where c is the speed of light (3.0×10^8 m/s) and h is Planck's constant (6.626×10^{-34} J·s). This equation shows that λ is also inversely proportional to ν (ie, longer wavelengths correspond to lower frequencies).

Therefore, in the given emission line spectrum of helium, the photon emission at 505 nm has a lower frequency than the photon emission at 470 nm.

(Choice A) The photon emission at 672 nm has a longer wavelength with a *lower* energy than the photon emission at 589 nm.

(Choice C) The energy of any photon emission is *directly proportional* to the photon frequency (ie, higher frequencies correspond to higher energies).

(Choice D) The energy of any photon emission is *inversely proportional* to the wavelength (ie, longer wavelengths correspond to lower energies).

Educational objective:

Each element emits photons only at specific wavelengths, giving a unique emission line spectrum corresponding to its electron configuration. The energy of a photon is directly proportional to its frequency and inversely proportional to its wavelength.

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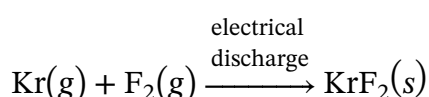
Subject:
General Chemistry

QID:403329

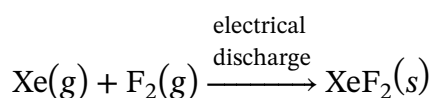
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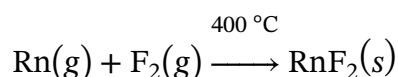


Reaction 1



Reaction 2

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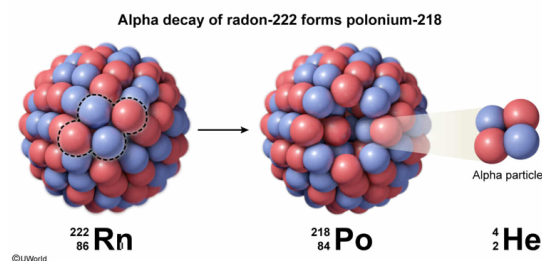
Atoms of radon-222 undergo radioactive decay and form polonium-218 by:

- ☐ A. the loss of 4 protons during positron emission. [8%]
- ☐ B. the loss of 4 neutrons during beta decay. [5%]
- ☐ C. the loss of 2 protons and 2 neutrons due to gamma emission. [7%]
- ☒ D. the loss of a helium-4 nucleus during alpha decay. [77%]

Correct

77% answered correctly

Explanation:



Radioactivity is the spontaneous emission of radiation (photons or particles) from an unstable atomic nucleus. Although several types of nuclear radiation exist, the three most common types are alpha decay, beta decay, and gamma emission.

During **alpha (α) decay**, an unstable nucleus ejects an α particle consisting of 2 protons and 2 neutrons (ie, a helium-4 nucleus without its electrons). Accordingly, the resulting nucleus formed following an α decay will have a **mass number** that is 4 units less and an **atomic number** that is 2 units less than the nucleus that underwent the α decay.

(Choices A and B) **Beta (β) decay** can occur in three forms: β^- decay (electron emission), β^+ decay (positron emission), and electron capture. In all three forms of beta decay, the *mass number is unchanged* but the atomic number either increases (β^- decay) or decreases (β^+ decay) by 1 unit.

(Choice C) Although **gamma emission** may sometimes accompany other radioactive decay processes, gamma emission itself emits only a high-energy photon (ie, no particle is emitted from the nucleus solely due to gamma emission). As such, the number of protons and neutrons in the nucleus remains constant and the elemental identity of the atom is unchanged.

Educational objective:

In alpha (α) decay, the mass number of an atomic nucleus decreases by 4 units and the atomic number decreases by 2 units. In all forms of beta (β) decay, the mass number of an atomic nucleus remains unchanged but the atomic number increases (β^- decay) or decreases (β^+ decay) by 1 unit. During gamma emission, only a high-energy photon is emitted, and the mass number and atomic number remain constant.

Time spent: 0 sec

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